

WEEK-3

Vector Space $\rightarrow$ is a set V with 2 operations (addition and scalar multiplication) satisfying all 8 properties above.

- Ex:
- $\mathbb{R}^n$ with usual addition and scalar multiplication
 - $M = \{m \times n\}(\mathbb{R})$ - set of all $m \times n$ real matrices
 - Polynomials of degree $\leq n$
 - Functions from $\mathbb{R}$ to $\mathbb{R}$

Q. If (a, b, c) is the inverse element of $(1, 2, 5) \in \mathbb{R}^3$ of the vector space $\mathbb{R}^3$, then find the value of $a - b + c$.

Sol. $(1, 2, 5) + (a, b, c) = (0, 0, 0)$
 $\Rightarrow (1, 2, 5) + (a, b, c) = (0, 0, 0)$

$$a = -1$$

$$b = -2$$

$$c = -5$$

Ans. $a - b + c = -4$

Subspace $\rightarrow$ Subset of a vector space V is a subspace if

1. Zero vector $\in W$
2. Closed under addition
3. Closed under scalar multiplication

$|A| = 0$ • Linear dependence $\rightarrow C_1 v_1 + C_2 v_2 + C_3 v_3 + \dots + C_n v_n = 0$

$|A| \neq 0$ • Linear Independence $\rightarrow C_1 = C_2 = C_3 = C_4 = \dots = C_n = 0$

Q. Consider 3 vectors $v_1 = (1, 0, 0)$, $v_2 = (1, 1, 0)$, $v_3 = (1, 1, 1)$ in the vector space $\mathbb{R}^3$, with usual addition and scalar multiplication. Which of the following sets are true?

Sol. $v_1 = (1, 0, 0)$, $v_2 = (1, 1, 0)$, $v_3 = (1, 1, 1)$

$$a_1(1, 0, 0) + a_2(1, 1, 0) + a_3(1, 1, 1) = 0$$

$$\Rightarrow (a_1 + a_2 + a_3, a_2 + a_3, a_3) = (0, 0, 0)$$

$$\Rightarrow a_1 + a_2 + a_3 = 0$$

$$a_2 + a_3 = 0$$

$$a_3 = 0$$

$\{v_1, v_2, v_3\}$ is LD $\times \rightarrow$ it's LI
 $\{v_1 + v_2, v_1 + v_3, v_3\}$ LD $\times \rightarrow$ it's LI
 $\{v_2 - v_1, v_1, v_3\}$ LD $\checkmark$

Q. Consider the set of matrices $S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ x \end{bmatrix} \right\}$
 $C M_{3 \times 1}(\mathbb{R})$. If $x \in \{9, 4, 20\}$, then find the maximum value of x so that S is linearly independent.

Sol.

$$A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & x \end{bmatrix}$$

$$|A| \neq 0$$

$$1(5x - 48) - 4(2x - 24) + 7(12 - 15) \neq 0$$

$$\Rightarrow -3x + 27 \neq 0$$

$$\Rightarrow x \neq 9$$

WEEK-4

Span $\rightarrow$ Span $\{[1, 0]\} \rightarrow$ all points $(x, 0)$

Span $\{[1, 0], [0, 1]\} \rightarrow \mathbb{R}^2$

Span $\{[1, 1], [2, 2]\} \rightarrow x = y$

So, generalising: Span $\{v_1, v_2, v_3, \dots, v_n\} = \{c_1 v_1 + c_2 v_2 + c_3 v_3 + \dots + c_n v_n\}$

Basis $\rightarrow$ Minimal set of independent vectors that still span the space
 $[1, 0], [0, 1]$ is a basis for $\mathbb{R}^2 \rightarrow$ dimension = 2

$\Rightarrow$ Q. Let S be the set of matrices where $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$
 P : Span (S) is the vector space consisting of only upper triangular matrix of order 2.

Proof $\rightarrow$

$$[S] = \{a_1 v_1 + a_2 v_2 + a_3 v_3\}$$

$$= a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$$

Q. Consider the following set of vectors

$$S = \{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$$

Sol. The set $\{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0)\}$ is not a basis of $\mathbb{R}^3$

set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$ is a basis of $\mathbb{R}^3$

$\Delta A = 0$ Dependent not a basis

$\Delta A \neq 0$

Q. If S be a subset of $\mathbb{R}^5$ such that $\text{span}(S) = \mathbb{R}^5$, then what is the minimum no. of elements possible in S ?

Sol

→ (5)

Dimension of a vector space

= no. of elements in any basis = cardinality of basis

= rank (maximal linearly independent set)

• $\dim(\mathbb{R}^n) = n$

• $\dim(M_{m \times n}(\mathbb{R})) = mn$

• $\dim(P_n) = n+1$ (polynomial of degree $\leq n$)

or
minimal spanning set

Q. Which of the following set forms a basis of vector space of 2×2 lower triangular matrices with usual matrix add/mult

Sol.

✓ $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\} \rightarrow a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$

✓ $\left\{ \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\} \rightarrow \begin{bmatrix} a & 0 \\ a+b & b \end{bmatrix}$ (lower triangular)

✗ $\left\{ \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$ ✗

$n-1$:
 $a \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} + c \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
 $= \begin{bmatrix} a+c & 0 \\ a+b & b+c \end{bmatrix}$

$n-2$:
 $\begin{bmatrix} a & 0 \\ b & c \end{bmatrix} \rightarrow (a, b, c)$

$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

$|A| = 2 \rightarrow$ Non-zero (linearly independent)

$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \Rightarrow (1, 1, 0)$

$\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \Rightarrow (0, 1, 1)$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow (1, 0, 1)$

Q. Find the dimension of the vector space $V = \{(x, y, z, w) \mid x+y = z+w, y+w = x+z, \text{ and } x, y, z, w \in \mathbb{R}\}$.

Sol.

$x+y = z+w$

$x+w = y+z$

after solving both eqn. $y=w$

$x=z$

Rank/dim = 2

- For finding rank correctly RREF form is preferred

WEEK-5

• Nullity = $n - \text{Rank}(A) =$ No. of independent variables for the system of equations $Ax=0$

Invertibility:

- $\text{Nul}(A) = \{0\}$, then matrix is invertible
- if $\text{Nul}(A)$ contains non-zero vector, A is not invertible because it collapses some part of the space.

- Scalar Matrix $\rightarrow$ Form: $A = \lambda I_n$

$$A = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

Q. Consider the matrix A given as follows:

$$A = \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 3 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Which of the following sets is a possible basis for the null space of A ?

Sol.

We need null space

$$Ax=0$$

① $x = (x_1, x_2, x_3, x_4)$

② write eqns:

1. $3x_1 + x_2 = 0$

2. $5x_2 = 0$

3. $3x_1 + 3x_3 = 0$

4. $0 = 0$ (ignoring)

$$\left. \begin{array}{l} 1. \quad 3x_1 + x_2 = 0 \\ 2. \quad 5x_2 = 0 \\ 3. \quad 3x_1 + 3x_3 = 0 \end{array} \right\} \begin{array}{l} \rightarrow x_2 = 0 \\ x_1 = 0 \\ x_3 = 0 \end{array}$$

x_4 not in any eqn. so, $x_4 = t$ (free)

Gen sol: $(x_1, x_2, x_3, x_4) = (0, 0, 0, t) = t(0, 0, 0, 1)$

③ $\text{Null}(A) = \text{span} \{ (0, 0, 0, 1) \}$

④

Basis must: • span null space
• be independent

So, $\{ (0, 0, 0, 1) \}$

Ex. 2

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

no contribution

Rank $\rightarrow 3$

Nullity $= 5 - 3 \Rightarrow 2$

Find null space $= Ax = 0$ let $x = (x_1, x_2, x_3, x_4, x_5)$

$\hookrightarrow x_2 = 0, x_1 = 0, x_5 = 0$

$x_3, x_4 \rightarrow \text{free}$

Gen sol: $x = (0, 0, x_3, x_4, 0)$

• Basis vectors: $\begin{cases} (0, 0, 1, 0, 0) \\ (0, 0, 0, 1, 0) \end{cases}$ spans the nullspace of A

Q. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a function for which the following is known: $T(1, 0) = (k, 1)$, $T(0, 1) = (1, k)$, $T(2, 1) = (-k^2 + k + 1, 2 - k^2)$, where $k \in \mathbb{R}$. Find the value of k for which T is a linear transformation which is not injective.

Sol.

• $T(1, 0) = (k, 1)$
• $T(0, 1) = (1, k)$
• $T(2, 1) = (-k^2 + k + 1, 2 - k^2)$

$T(2, 1) = 2T(1, 0) + T(0, 1)$
 $= 2(k, 1) + (1, k) = (2k + 1, 2 + k)$

$\Rightarrow (2k + 1, 2 + k) = (-k^2 + k + 1, 2 - k^2)$

$k = 0 \text{ or } k = -1$

3

Check non-injectivity

A linear transformation

is not injective $\Leftrightarrow$

its matrix is $n \times n$
invertible $\Leftrightarrow |A| \neq 0$

Matrix of T :

$$A = \begin{bmatrix} k & 1 \\ 1 & k \end{bmatrix}$$

$$|A| = k^2 - 1$$

$$k^2 - 1 = 0$$

$$k = \pm 1$$

Ans. • common value $k = -1$

$$\boxed{W \in E_{k-6}}$$

1. linear transformations:

$$T(u+v) = T(u) + T(v) \quad (\text{additivity})$$
$$T(cu) = cT(u) \quad (\text{homogeneity})$$